WIFI Audio
HTTP API

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1. Brief

1.1. brief

Wifi Audio AXX module is the SoC module for WiFi audio solutions, it support Smartlink, DLNA and Airplay.
It also support some http API for quick access.

1.2. HTTP get

You can send HTTP get request to the module, the response if in JSON.

Request format is http://x.x.x.x/httpapi.asp?command=********
X.x.x.x is the IP address, (Below, assume the IP is 10.10.10.254)
******* is the command.

1.3. About Hexed

Some data should hexed before transfer it.
Here is the hex method (in C):

```
int hex2ascii(const char *pSrc, unsigned char *pDst, unsigned int nSrcLength, unsigned int nDstLength)
{
	int i, j = 0;
	memset(pDst, 0, nDstLength);
	for (i = 0; i<nSrcLength; i+=2 )
	{
		char val1 = pSrc[i];
		char val2 = pSrc[i+1];
		if( val1 > 0x60) val1 -= 0x57;
		else if(val1 > 0x40) val1 -= 0x37;
		else val1 -= 0x30;
		if( val2 > 0x60) val2 -= 0x57;
		else if(val2 > 0x40) val2 -= 0x37;
		else val2 -= 0x30;
		if(val1 > 15 || val2 > 15 || val1 < 0 || val2 < 0)
			return 0;
		pDst[j] = val1*16 + val2;
		j++;
	}
	return j;
```

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```
}

int ascii2hex(char* ascii_in, char* hex_out, int ascii_len, int hex_len)
{
	const char hex[16] = {'0', '1', '2', '3', '4', '5', '6', '7', '8','9', 'A', 'B', 'C', 'D', 'E', 'F'};
	int i = 0;
	int ret = 0;
	memset(hex_out, 0, hex_len);
	while( i < ascii_len )
	{
		int b= ascii_in[i] & 0x000000ff;
		hex_out[i*2] = hex[b/16] ;
		hex_out[i*2+1] = hex[b%16] ;
		++i;
		ret += 2;
	}
	return ret;
}
```

2. API list

2.1. Get device information

http:// 10.10.10.254/httpapi.asp?command=getStatusEx

- result：


```
{
	"language": "en_us",
	"ssid": "FA5100_a4dc",
	"firmware": "WIFIAudio.multil_room.1.2.20140324 ",
	"builddate": "release ",
	"Release": "20140324 ",
	"group": "",
	"expired": "0",
	"internet": "0",
	"uuid": "de cf 1d 2e b1 60 e4 38 ",
```


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```
    "netstat": "0",
    "essid": "",
    "apcli0": "",
    "eth2": "192.168.120.112",
    "hardware": "WiiMu-A03 ",
}
```


language	language
ssid	SSID
firmware	version
builddate	Can be: release, debug, backup Release: means this is a release version Debug: means this is a debug version Backup means this is a backup version
Release	The build date
group	
expired	1 means the firmware has expired
internet	Internet access: 0: not ready; 1: ready
uuid	UUID
netstat	WIFI Station connect state: 0: no connection; 1：connecting; 2：connected
essid	The router name connected. The name is hexed
apcli0	WIFI Station IP address
ra0	WiFi AP IP address, normally is 10.10.10.254
eth2	Ethernet IP address
hardware	The hardware version
project	The project name
VersionUpdate	Is there new version.
NewVer	If there is new version, this is the version number.
DeviceName	The device upnp and airplay friendly name
temp_uuid	Temp uuid, will change after boot up
capability	Bit: 0 if airplay support 1 if ethernet support 2 if USBdisk support 3 if WPS button support

	4 if battery support 5 if preset key support 6 if I2S in support
streams	Bit: 0 if airplay enabled 1 if DLNA enabled 2 if TTPod support 3 if TuneIn support 4 if Pandora support 5 if DoubanFM support
external	
preset_key	presets key number
plm_support	Bit: 0 LineIn (Aux support) 1 BT support 2 optical support
WifiChannel	WiFi channel
AP_clients	AP client number
RSSI	Router RSSI (0 ~ 100), 100 is the best
TxQuality	Tx quality,
RxQuality	Rx quality
battery	1 means battery is in charging
battery_percent	0 ~100
securemode	WiFi secure or not
psk	If WiFi is in secure, the password
usb_storagesize	Udisk storage size
usb_freesize	Udisk free size
part1_storage	User1 part in ROM, storage size
part1_free	User1 part in ROM, free size
part2_storage	User2 part in ROM, storage size
part2_free	User2 part in ROM, free size

2.2 Network

2.2.1 Get the scan AP list

http:// 10.10.10.254/httpapi.asp?command=wlanGetApListEx

{"res":"0","aplist":[{"ssid":"wmmAudio_a7b8","bssid":"00:22:6c:00:a7:b8","rssi":"76","channel":"1","auth":"OPEN","encry":"NONE","extch":"1"},{"ssid":"WIIMU_Network","bssid":"20:dc:e6:cb:7e:78","rssi":"70","channel":"11","auth":"WPA2PSK","encry":"AES","extch":"0"},{"ssid":"WIFIAudio_29b0","bssid":"00:22:6c:16:29:b0","rssi":"65","channel":"11","auth":"OPEN","encry":"NONE","extch":"0"}]}

res	number
aplist	
ssid	Wifi SSID name (hide SSID is not include), hexed
bssid	MAC
rssi	RSSI (0~100)
channel	Wifi channel
auth	Authorization
encry	Encrypt
extch	

2.2.2 Connect wifi

http://10.10.10.254/httpapi.asp?command=wlanConnectApEx:ssid=xxx:ch=n:auth=xxx:encty=xxx:pwd=xxx:chext=n

Connect the device to a router. When connecting, the device AP connection may lost.

This API has no return result, you can call wlanGetConnectState to get the connect state if the AP connection still not lost.

ssid	Router ssid，hexed
channel	Wifi channel
auth	Authorization
encry	encrypt
pwd	Passowrd, hexed，if no password, not input here
chext	1

2.2.3 connect hide wifi

http://10.10.10.254/httpapi.asp?command=wlanConnectHideApEx:ssid:pwd

Connect the device to a hide router. When connecting, the device AP connection may lost.

This API has no return result, you can call wlanGetConnectState to get the connect state if the AP connection still not lost.

Ssid and pwd must hexed， if no pwd， API is:

http://10.10.10.254/httpapi.asp?command=wlanConnectHideApEx:ssid

2.2.4 Get the connect state

http://10.10.10.254/httpapi.asp?command=wlanGetConnectState

Note the return result is not in json

- return：

返回值字段	字段说明
PROCESS	In progress
PAIRFAIL	Wrong password
FAIL	Connect fail
OK	connected

2.3 Playback control

Note, the 3rd party DLNA can't be controlled.

2.3.1 Get playback state

http://10.10.10.254/httpapi.asp?command=getPlayerStatus

```
{
    "mainmode": "0",
    "nodetype": "0",
    "mode": "3",
```

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```
    "sw": "0",
    "status": "play",
    "curpos": "12900",
    "totlen": "229000",
    "Title": "736865",
    "Artist": "47726f6f766520436f766572616765",
    "Album": "xxxxxxxxxx",
    "Year": "2005",
    "Track": "7",
    "Genre": "Dance",
    "locallistflag": "1",
    "locallistfile": "",
    "plicount": "1",
    "plicurr": "1",
    "vol": "90",
    "mute": "0",
    "iuri": "xxxxxxxxxxxxxxxxxxxxx",
    "uri": "xxxxxxxxxxxxxxxxxxxxxx"
}
```


- 字段说明：

返回值字段	字段说明
type	0: master or standalone device 1: slave
ch	0 stereo，1 left，2 right
mode	0 none 1 Airplay 2 3rd party DLNA 11 ~ 19 Wiimu playlist, 11 is the USB disk playlist 21 ~ 29 Http API playback, 21 is the USB disk playback 30 alarm 40 AUX 41 BT 42 external storage 43 optical 50 mirror

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	60 voice mail 99 slave
loop	Loop mode: 0: sequence, no loop 1: single loop 2: shuffle all 3: loop all
eq	The Equalizer number
status	0 stop 1 play 2 load 3 pause
curpos	Position, in ms
totlen	Duration in ms
Title	Return hexed data, the music name or music URL
Artist	Return hexed data
Album	Return hexed data
udsikflag	Is there USB disk
plicount	The track number of playlist
plicurr	Current track index
vol	Current volume
mute	Current mute state

2.3.2 playback

http://10.10.10.254/httpapi.asp?command=setPlayerCmd:play:uri

Play the URI

http://10.10.10.254/httpapi.asp?command=setPlayerCmd:playlist:uri:<index>

Play the URI (URI is the m3u playlist, index is the start index)

http://10.10.10.254/httpapi.asp?command=setPlayerCmd:hex_playlist:uri:<index>

Play the URI (URI is the m3u playlist, index is the start index), here, uri is hexed

http://10.10.10.254/httpapi.asp?command=setPlayerCmd:playLocalList:<index>

Play the USB disk, index is the start index

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2.3.3 Pause / Resume

http://10.10.10.254/httpapi.asp?command=setPlayerCmd:pause

http://10.10.10.254/httpapi.asp?command=setPlayerCmd:resume

2.3.4 Pause or Resume

http://10.10.10.254/httpapi.asp?command=setPlayerCmd:onepause

If the state is paused, resume it; or, pause it.

2.3.5 Previous

http://10.10.10.254/httpapi.asp?command=setPlayerCmd:prev

2.3.6 Next

http://10.10.10.254/httpapi.asp?command=setPlayerCmd:next

2.3.7 Seek

http://10.10.10.254/httpapi.asp?command=setPlayerCmd:seek:position

Position is from 0 to duration in second

2.3.8 stop

http:// 10.10.10.254/httpapi.asp?command=setPlayerCmd:stop

2.3.9 Set volume

Vol++
http://X.X.X.X/httpapi.asp?command=setPlayerCmd:Vol%2B%2Bn
Vol--
http://X.X.X.X/httpapi.asp?command=setPlayerCmd:Vol--n

to show the playing volume
http://X.X.X.X/httpapi.asp?command=setPlayerCmd:Vol--n

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note: n is from 1-100, it stands each volume tuning will increase or decrease the volume by n%.

2.3.10 Mute

http://10.10.10.254/httpapi.asp?command=setPlayerCmd:mute:n

```
Mute: n=1
Unmute: n=0
the slave mute state will be set at the same time.
```

2.3.11 Loop mode set

http://10.10.10.254/httpapi.asp?command=setPlayerCmd:loopmode:n

```
n
 0                        Sequence, no loop
 1                        Single loop
 2                        Shuffle loop
 -1                       Sequence loop
```

2.3.12 EQ set

http://10.10.10.254/httpapi.asp?command=setPlayerCmd:equalizer:mode

```
mode
 0                        disable
 1                        Classic
 2                        Popular
 3                        Jazzy
 4                        Vocal

Not all projects support EQ.
```

2.3.13 EQ get

http://10.10.10.254/httpapi.asp?command=getEqualizer

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2.4 USB disk playback

2.4.1 Get the playlist of USB

http://10.10.10.254/httpapi.asp?command=getLocalPlayList


```
{
    "num": "2",
    "locallist": [
        {
            "file": "/media/sda1/avril lavigne - tik tok.mp3"
        },
        {
            "file": "/media/sda1/avril lavigne – hush hush.mp3"
        }

    ]
}
```


num	Music file number in USB disk
locallist	
file	Music file，hexed

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2.4.2 Get Music information in USB disk

http:// 10.10.10.254/httpapi.asp?command=getFileInfo:index:range

Get the music metadata，index is the sequence number in USBDisk playlist, range is the number

If range is large than 1， return multiple fileinfo

For example:

{“num”:”2”, “infolist”:[

{
"filename": "/media/sda1/avril lavigne – hush hush.mp3",
"totlen": "0",
"Title": "unknown",
"Artist": "unknown",
"Album": "unknown"
},
{
"filename": "/media/sda1/avril lavigne – hush hush2.mp3",
"totlen": "0",
"Title": "unknown",
"Artist": "unknown",
"Album": "unknown"
}]}

Or return single metadata

{
"filename": "/media/sda1/avril lavigne – hush hush.mp3",
"totlen": "0",
"Title": "unknown",
"Artist": "unknown",
"Album": "unknown"
}

- 字段说明：

返回值字段	字段说明
filename	Filename, hexed
totlen	
Title	Return unknown or hexed data
Artist	Return unknown or hexed data
Album	Return unknown or hexed data

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2.4.3 Play the USBDisk

http:// 10.10.10.254/httpapi.asp?command=setPlayerCmd:playLocalList:index

2.6 Multiroom

2.6.1 Get slave list

http://10.10.10.254/httpapi.asp?command=multiroom:getSlaveList

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```
{
    "slaves": "1",
    "slave_list": [
        {
            "name": "FA5100_a3f4",
            "mask": "0",
            "volume": "90",
            "mute": "0",
            "channel": "0",
            "ip": "10.10.10.100",
            "version": "WIFIAudio.1.2.2321"
        }
    ]
}
```


slaves	Slave number
slave_list	
name	Slave device name
mask	If slave is masked, it will act as a standalone device.
Volume	Slave volume
Mute	slave is mute or not
Channel	Slave channel: 0 stereo 1 left 2 right
ip	Slave IP address
version	Slave firmware version
Ssid	Slave ssid
Uuid	Slave uuid

2.6.2 2.6.3 kickout slave

http://10.10.10.254/httpapi.asp?command=multiroom:SlaveKickout:ip

Kickout one slave from group.

2.6.4 mask slave

http://10.10.10.254/httpapi.asp?command=multiroom:SlaveMask:ip

Mask one slave, If slave is masked, it will act as a standalone device.

2.6.5 un-mask slave

http://10.10.10.254/httpapi.asp?command=multiroom:SlaveUnMask:ip

2.6.6 set slave volume

- 接口 URL：

http://10.10.10.254/httpapi.asp?command=multiroom:SlaveVolume:ip:volume

2.6.7 set master volume

- 接口 URL：

http://10.10.10.254/httpapi.asp?command=setPlayerCmd:slave_vol:volume

Just set the master or standalone volume, not affect the slave volume.

2.6.8 mute slave

http://10.10.10.254/httpapi.asp?command=multiroom:SlaveMute:ip:mute

2.6.9 mute master

http://10.10.10.254/httpapi.asp?command=setPlayerCmd:slave_mute:mute

Just mute the master or standalone , not affect the slave .

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2.6.10 Slave playback channel set

http://10.10.10.254/httpapi.asp?command=multiroom:SlaveChannel:ip:channel

Slave channel:
0 stereo
1 left
2 right

2.6.11 Master playback channel set

http://10.10.10.254/httpapi.asp?command=setPlayerCmd:slave_channel:channel

Just set the master or standalone playback channel , not affect the slave .
Slave channel:
0 stereo
1 left
2 right

2.6.12 start WPS server

http://10.10.10.254/httpapi.asp?command=wpsservermode

Start WPS server, wait for the slave connection

2.6.13 stop WPS server

http://10.10.10.254/httpapi.asp?command=wpscancel

2.6.14 start WPS Client

http://10.10.10.254/httpapi.asp?command=wpsclientmode

scan the WPS server and connect it

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2.8 Device control

2.8.1 Set SSID

http:// 10.10.10.254/httpapi.asp?command=setSSID:value

Set the device SSID
SSID value should < 16 bytes without special charaters.

2.8.2 Set AP password

http:// 10.10.10.254/httpapi.asp?command=setNetwork:1:password

Password value should < 16 bytes without special charaters.

http:// 10.10.10.254/httpapi.asp?command=setNetwork:0

Remove AP passowrd

2.8.3 Restore

http:// 10.10.10.254/httpapi.asp?command=restoreToDefault

2.8.4 Reboot

http:// 10.10.10.254/httpapi.asp?command=reboot

2.8.5 shutdown

http:// 10.10.10.254/httpapi.asp?command=setShutdown:sec

Shutdown device in sec

```
sec:
    0: shutdown immediately
    -1: cancel the previous shutdown timer
```

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2.8.6 Get the shutdown timer

http:// 10.10.10.254/httpapi.asp?command=getShutdown

Return the seconds

2.8.7 Close WIFI

http:// 10.10.10.254/httpapi.asp?command=setPowerWifiDown

Device is still active but Wifi is closed.

2.9 Online Upgrade

2.9.1 Check the new version

http:// 10.10.10.254/httpapi.asp?command=getMvRemoteUpdateStartCheck

2.9.2 Start upgrade

http:// 10.10.10.254/httpapi.asp?command=getMvRemoteUpdateStart

It will download the firmware first and then burn it.

2.9.3 Check the upgrade status

http:// 10.10.10.254/httpapi.asp?command=getMvRemoteUpdateStatus

return	
40	Find new version
10	checking
21	Check fail
22	Download fail
23	Verify firmware fail
25	downloading
27	Download finished, burning (for user part)
30	Download finished, burning
others	No new version

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2.9.6 Get the upgrade percent

http:// 10.10.10.254/httpapi.asp?command=getMvRomBurnPrecent

```
{"status":"0","progress":"50"}
```

status	0 in progress -1 failed
progress	0~100

2.10 Alarm clock

2.10.1 time sync

If the device has no internet access, you need to sync its time with:
http://10.10.10.254/httpapi.asp?command=timeSync:YYYYMMDDHHMMSS

YYYY is year (such as 2015), MM is month (01~12), DD is day (01~31), HH is hour (00~23), MM is minute (00~59), SS is second (00~59)

In UTC

2.10.2 alarm set

http://10.10.10.254/httpapi.asp?command=setAlarmClock:n:trig:op:time[:day][:url]

n: 0~2, currently support max 3 alarm
Trig: the alarm trigger:
0 cancel the alarm, (AXX+TMR+S010&)
1 once, day should be YYYYMMDD
2 every day, day is no need
3 every week, day should be 2 bytes (00”~“06”), means from Sunday to Saturday.
4 every week, day should be 2 bytes, the bit 0 to bit 6 means the effect, for example, “7F” means every day in week, “01” means only Sunday。
5 every month, day should be 2 bytes (“01”~“31”)

op: the action
0 shell execute

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1 playback or ring
2 stop playback

Time should be HHMMSS

url: the shell path or playback url, should less than 256 bytes

2.10.3 Get alarm

http://10.10.10.254/httpapi.asp?command=getAlarmClock:n

n: 0~2, currently support max 3 alarm

{"enable":"1",

"trigger":"%d",

"operation":"%d",

"date"::"%02d:%02d:%02d", //if not a “every day” alarm, no this

"week_day":"%d", //if not a “every week” alarm, no this

"day":"%02d", //if not a “every month” alarm, no this

"time":"%02d:02d:%02d",

"path":"%s""}

2.10.4 Stop the current alarm

http://10.10.10.254/httpapi.asp?command=alarmStop

2.11 Playback source

2.11.1 switch playback source

http://10.10.10.254/httpapi.asp?command=setPlayerCmd:switchmode:%s

line-in:

bluetooth:

optical:

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Above, device will capture I2S data and play it

udisk：if Udisk is connected to the device, it will playback the UDisk, or device will capture I2S data and play it

wifi：

2.12 GPIO simulation

2.12.1 Pull up GPIO

http://10.10.10.254/httpapi.asp?command=IOSimuPullUp:%d

00: GPIO_nReload
11: GPIO_nReady （only A02）
12: GPIO_DCD
14: GPIO_RIN
18: GPIO1（only A11）
17: GPIO2（only A11）

2.12.2 Pull down GPIO

http://10.10.10.254/httpapi.asp?command=IOSimuPullUp:%d

2.12.3 Read GPIO level

http://10.10.10.254/httpapi.asp?command=IOSimuRead:%d

2.12.4 Key simulation

http://10.10.10.254/httpapi.asp?command=IOSimuKeyIn:%d

Simulate the MCU+KEY+XXX command

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http://10.10.10.254/httpapi.asp?command=IOSimuKeyOut:%d

Generate the AXX+KEY+XXX command to MCU

2.13 Voice prompt

http://10.10.10.254/httpapi.asp?command=PromptEnable
http://10.10.10.254/httpapi.asp?command=PromptDisable

2.14 pre-set program 1-6

http://10.10.10.254/httpapi.asp?command=MCUKeyShortClick:n
n is from 1-6

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